ILLANGHKUMARAN THIRUVALLUVAN

Platform Engineer / Site Reliability Engineer (SRE)

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PROFESSIONAL SUMMARY

Results-driven Platform & Site Reliability Engineer with 6+ years of enterprise experience at Tata Consultancy Services. Specialized in cloud-agnostic distributed systems engineering, end-to-end infrastructure automation (IaC), and high-availability container orchestration (Kubernetes). Leverages an M.S. in Software Engineering to engineer resilient system architecture, reducing deployment overhead by 87.5%, optimization of production cloud spend by 15%, and eliminating operational downtime across business-critical workflows matching Tier-1 engineering baselines.

TECHNICAL SKILLS

- **Container Orchestration & Platforms:** Kubernetes (AKS/GKE), Docker, App Services, Virtual Networks, Cluster Auto-scaling
- **Infrastructure as Code & CI/CD:** Terraform, GitOps Architecture, GitHub Actions, Azure DevOps (Pipelines, Repos), Git
- **Systems Programming & Scripting:** Python (Systems Automation & API integration), PowerShell, Bash, Helm Charts
- **Networking & Traffic Engineering:** Cloudflare WAF, Enterprise DNS/Edge Routing, Venafi/Sectigo PKI
- **Observability & Telemetry:** Grafana, Prometheus, Azure Monitor, Log Analytics, Application Insights, Elasticsearch
- **Operating Systems & Systems Data:** Windows Server (2012-2025), Linux (RHEL 7, 8)

PROFESSIONAL EXPERIENCE

Tata Consultancy Services – *Platform & Infrastructure Automation Engineer* | Feb 2020 – Present | Chennai, India

- **Cloud Infrastructure Engineering:** Designed and provisioned 50+ modular enterprise multi-tier cloud environments utilizing reusable, cloud-agnostic Terraform blueprints, accelerating infrastructure orchestration delivery times by 75%.
- **Container & Cluster Orchestration:** Spearheaded the architectural migration of 15+ legacy on-premises applications to managed Kubernetes clusters; engineered dynamic Horizontal Pod Autoscaler (HPA) baselines sustaining 200% traffic spikes with zero service degradation while accelerating API response times by 20%.
- **Continuous Delivery & Pipeline Design:** Automated resilient end-to-end YAML-based declarative CI/CD engines for microservices architectures, compressing deployment lifecycles from 2 hours down to 15 minutes (an 87.5% operational reduction).
- **GitOps & Production Governance:** Enforced strict, pull-request-driven GitOps workflows for systemic state changes, mitigating structural configuration anomalies and runtime environment drifts by 40%.
- **Security & Identity Automation:** Standardized zero-trust infrastructure paradigms by engineering automated SSL/TLS public key cryptography lifecycle rotations (Venafi/Sectigo API wrappers), mitigating 100% of expiry-induced system outages.
- **Traffic & Edge Engineering:** Built a cross-platform integration engine unifying edge routing states between legacy F5 load balancers and Cloudflare Enterprise DNS setups, automating away 80% of routine technical interventions.

- **High Availability & SRE:** Managed multi-region cluster deployment topologies incorporating automated cross-region failover protocols, consistently meeting an enterprise-grade 99.95% global system availability SLA.
- **Telemetry & Auto-Remediation:** Co-authored intelligent automated webhook handler scripts (Python/PowerShell) bound to Grafana and centralized alerting layers, dropping production Mean Time to Resolution (MTTR) by 30%.

KEY PROJECTS

Enterprise Edge & Domain Migration to Cloudflare

- Orchestrated and navigated the massive migration of 2,000+ business-critical root domains from legacy on-premises F5 nameservers to Cloudflare Enterprise, utilizing automated pipeline scripts.
- Executed global cutover workflows with absolute zero down-time or business disruption, resulting in a 95% performance speedup in global edge routing and DNS propagation latencies.

Automated Edge Zone Provisioning Engine

- Coded and compiled an automated DNS lifecycle configuration manager by writing custom API integration modules directly inside native infrastructure pipelines.
- Reclaimed over 30+ engineering hours per month by transitioning configuration work to deterministic, automated validation frameworks.

Automated Server Deployment via Reusable Cloud Architecture (Terraform)

- Developed and published an enterprise repository of modular, highly flexible Terraform definitions to automate multi-tier networking and compute stack configurations (VNETs, Subnets, Firewalls, VM scale configurations).
- Decoupled secrets management by integrating native cloud secrets vaults dynamically into running deployment pipelines, validating automated plans safely prior to runtime execution.
- Dropped environmental infrastructure target build delivery windows from days down to under 30 minutes while enforcing 100% compliance alignment against security policies.

Automated IIS Web Server Lifecycle Management System

- Engineered a production-ready PowerShell automation module leveraging WebAdministration controllers to programmatically provision, modify, and teardown multi-tenant IIS websites and application pools.
- Designed idempotent setup-and-deletion handlers that auto-verify system dependencies, bind SSL/TLS host headers, and cleanly recycle directory resources, eliminating 95% of legacy manual administration drift on Windows Server instances.

CERTIFICATIONS

- Microsoft Certified: Designing and Implementing Microsoft DevOps Solutions (AZ-400) - 2025
- Microsoft Certified: Azure Administrator Associate (AZ-104) - 2024
- Microsoft Certified: Power BI Data Analyst Associate (PL-300) – 2025
- Microsoft Certified: Azure AI Engineer Associate – (AI -200) - 2025
- Advanced Certification: Windows Server Administrator - 2026
- HashiCorp: Certified Terraform Associate - (In Progress)

EDUCATION

Vellore Institute of Technology (VIT) – *M.S. in Software Engineering* | Graduated: 2019 | Vellore, India

- **Thesis:** "User Product Review Analysis" (Data Mining, Sentiment Analysis, and Naïve Bayes text structures)

AWARDS & RECOGNITION

- **Top Performer of the Year (2024-2025):** Awarded by TCS for outstanding systemic cloud infrastructure deliveries.
- **Applause Award (2022):** Received for structural cloud cost optimizations and infrastructure scale design.
- **Employee of the Month (5x Recipient):** Awarded across 2022-2026 for incident resolution excellence and production stability.